Wirtschaftsuniversität Wien
Vienna University of Economics and Business

List of Aids for Seminar Paper/Thesis

Title of Paper/Thesis:
Coordinated Malicious Behavior Detection in Blockchain Networks Using Temporal Graph Motifs

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Author(s):
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Aids/Tools Used:

Aids/Tools Used	**Type(s) of Use**	**Affected Areas/Chapters**	**Documentation**
Claude	Converting prose to LaTeX; generating tables; writing Python code and building the analysis tool.	Individual passages and tables throughout; implementation in Chapter 4 and the appendix; analyses for Chapter 5.	Thesis LaTeX source and Python project source code.
ChatGPT	Converting prose to LaTeX and generating tables.	Individual passages and tables throughout.	Thesis LaTeX source and table files.
Perplexity	Converting prose to LaTeX and generating tables.	Individual passages and tables throughout.	Thesis LaTeX source and table files.
Overleaf writing tools	Writing and typesetting; language corrections and suggestions for synonyms.	Document preparation and individual passages throughout.	Overleaf project and thesis LaTeX source.
Codex	Final language, formatting and bibliography review; corrections and this list of aids.	Final thesis revision, bibliography and list of aids.	Review conversation, revised files and review report.

Aids/Tools Used (continued):

Aids/Tools Used	**Type(s) of Use**	**Affected Areas/Chapters**	**Documentation**
Python and Raphtory	Analysis pipeline, temporal graphs and motif-role counting.	Chapters 4–5 and implementation appendix.	Section 4.7, Table 4.5; source code and run protocols.
DuckDB, PyArrow, pandas, NumPy, SQLite	Data loading, preparation and storage; numerical calculations and feature construction.	Data preparation in Chapter 4; analyses in Chapter 5.	Section 4.7, Table 4.5; source code and run protocols.
scikit-learn, XGBoost	Data splitting, scaling, classifier training and evaluation.	Methods in Chapter 4; results in Chapter 5.	Sections 4.5–4.7; source code and run protocols.
Matplotlib	Pipeline diagrams and plots of experimental results.	Figures in Chapters 4–5.	Section 4.7; plotting scripts and figure files.
NetworkX, Hugging Face datasets	Exploratory graph analysis and dataset downloading.	Exploration and data acquisition in Chapter 4.	Section 4.7 and project source code.

I hereby declare that I have listed all the aids I have used in the list above. If no aids have been used, it is also indicated in the list (to be listed under *Aids/Tools Used: none*).

Date: **Signature:**